TKO 7093 Statistical Data Analysis

Time-Use Habits in Finland

A Statistical Analysis of the Time Use Survey

Group: Group 004

Course: TKO 7093 Statistical Data Analysis

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AI-use declaration. Claude (Anthropic) was used as a preliminary roadmap to help organize our workflow step-by-step and as a double-check tool during the analysis. Additionally, it assisted in organizing and structuring the layout and formatting in Overleaf. However, all methodological choices, logic, and analytical decisions were guided and driven by us. All quantitative results derive directly from our executed notebook, and this report was drafted with our own criteria. Appendix A describes every interaction.

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1 Introduction

1.1 Scenario and research questions

A research organisation commissioned a survey about how individuals and households in Finland spend their time. The Head of Research asked our team to analyse the data and to answer four questions:

1. Characterise the individuals that are present in the data.
2. Estimate how much time on average Finnish households spend daily on each activity.
3. With respect to which activities do living environments or days of week differ in Finland?
4. Which activities are associated with each other in Finnish population?

Each section explains what we wanted to find out, which method we used and why, and what we found. The conclusions at the end answer the four questions

1.2 The data

The data comes from the Time Use Survey of Statistics Finland (teaching version) and is released under the Creative Commons Attribution 4.0 licence. It comes as two files: `habits.data` with the observations and `habits.txt` with the documentation. Each row describes one person on one diary day. Table 1 lists the variables.

Table 1: Variables in the data, as documented in `habits.txt`.

Variable	Meaning	Coded values
`kohde`	Household ID	integer
`jasen`	Member ID within household	integer
`pvknro`	Day of the diary	1 = working day, 2 = weekend
`sp`	Sex	1 = male, 2 = female
`IKAL1`	Age	documented as a 9-level age group, but see Section 3.3
`ASALUE`	Living environment	1 = city, 2 = municipality, 3 = rural area
`A1`	Working	minutes
`A2`	Sleeping	minutes
`A3`	Reading	minutes
`A4`	Dining at a restaurant	minutes
`A5`	Visiting a library in the past 12 months	1 = yes, 2 = no

`A1` to `A4` are continuous time variables, and `A5` is a yes/no variable. Keeping this distinction in mind is essential when selecting appropriate techniques, whether we are running hypothesis tests, correlation matrices, or PCA.

1.3 Code and reproducibility

All results come from our Python notebook, which you can check along with the dataset and code in our GitHub repository: GitHub repository

1.4 How the report is organised

The report follows the following structure. Section 2 gives a short statistical background. Section 3 describes data preparation. Section 4 explores the data with descriptive statistics and a PCA. Section 5 estimates average daily times. Sections 6 and 7 contain the statistical tests, and Section 8 answers the four questions. Appendix A documents our use of AI.

2 Statistical background

This section explains, in plain terms, the methods used later. We describe them once here and refer back to them in the analysis.

Confidence interval for a mean. For a sample of size n with mean $\bar{x}$ and standard deviation s, a 95% confidence interval for the population mean is

$$\bar{x} \pm t_{0.975,\,n-1}\,\frac{s}{\sqrt{n}}.$$

We use the t-distribution because the population variance is unknown and has to be estimated. By the Central Limit Theorem, the sample mean is approximately normal when n is large, so the interval is reasonable even for skewed variables. In our data n is about 370 in each group.

Independence. Confidence intervals and most tests assume that observations are independent. In our data a person appears on up to two rows (one working day and one weekend day). If we treated those two rows as two different people, we would act as if we had more information than we do, and our intervals would be too narrow. Section 3.8 explains how we avoid this in every analysis.

Normality check and paired test. To compare a working day with a weekend day for the same people, we look at the paired differences $d_i = x_{i,\text{working}} - x_{i,\text{weekend}}$. The Shapiro-Wilk test checks whether these differences look normal. If they do, the paired t-test is used. If not, we use the Wilcoxon signed-rank test, which ranks the absolute differences $|d_i|$ and checks whether positive and negative differences balance out around zero.

Comparing two independent groups. To compare a continuous variable between two independent groups (e.g., people who visited a library and people who did not), we use the unpaired t-test if the data are approximately normal, and the Mann-Whitney U test otherwise.

Comparing three or more groups. To compare three groups (city, municipality, rural area), we use ANOVA if data are normal, or the Kruskal-Wallis test otherwise. Kruskal-Wallis extends the Mann-Whitney approach to multiple groups.

Test for two categorical variables. Pearson's chi-squared test compares the observed counts O_{ij} in a contingency table with the counts $E_{ij} = r_i c_j / n$ expected if the two variables were independent:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

It needs the expected counts not to be too small (a common rule is at least 5).

Correlation analysis. While Pearson's correlation assumes linear relationships and normality, **Spearman's correlation (ρ)** is the Pearson correlation of the data ranks. It measures monotonic association, is robust to skew and outliers, and handles ties with average ranks. This perfectly suits survey data, where many people report zero minutes.

Multiple testing. When we run m tests at once, the chance that at least one looks significant by luck grows quickly. With a 5% threshold, four independent tests already give more than an 18% chance of at least one false alarm. We control this with the False Discovery Rate (FDR) method of Benjamini and Hochberg. If the sorted p-values are $p_{(1)} \leq \cdots \leq p_{(m)}$, the adjusted value is

$$p_{\text{fdr}}^{(i)} = \min_{j \geq i} \frac{m}{j} p_{(j)}.$$

We apply FDR separately to each group of tests that answer the same question, and we always decide using the adjusted p-value p_{fdr}.

Principal Component Analysis (PCA). PCA looks for new axes (components) that capture as much of the variation in the data as possible. We first standardise each variable to mean 0 and variance 1, so that variables with large variances do not dominate. The components then come from the eigenvectors of the correlation matrix, and each eigenvalue is the variance carried by its component. The Kaiser criterion keeps the components with an eigenvalue above 1, that is, components that explain more than one original standardised variable.

3 Data preparation

The raw text file has to become a clean, correctly typed table before any statistics make sense. We start by reading the documentation and checking what a row really represents. Then we parse the time variables, sort out the age and library variables, deal with missing values and look for impossible values. We explain each decision as we go.

3.1 What does a row represent?

The documentation calls `kohde` a household ID and `jasen` a member ID. Before aggregating anything, we checked two things: how many people live in each household, and how many diary

days each person has. The file has 745 rows and 11 columns.

Table 2: Structure of the raw data.

Check	Result
Rows in the file	745
Distinct households (`kohde`)	378
Households with exactly one member	378
People with both diary days	367
People with only one diary day	11
Rows on working days	377
Rows on weekend days	368

Every household has exactly one sampled member, and each person contributes up to two rows, one per day type. So a row is one person on one specific day, and "household" and "individual" mean the same thing here. We still wrote the code to sum activity times by household and day, so that it keeps working if a larger version of the survey has households with two or more people.

3.2 Parsing the time variables

A first look at the raw values showed that the same column mixes plain numbers (for example `380`) with `HH:MM` strings (for example `01:20`). The code `?` marks a missing value. We converted everything to minutes: a value with a colon becomes hours $\times$ 60 + minutes, a plain number is already in minutes, and `?` becomes missing.

To check the parser, we looked at the resulting ranges and at the daily total of the four activities. A total above 1440 minutes (24 hours) is physically impossible and would point to a data error. A total below 1440 is normal, because the survey does not cover the whole day.

Table 3: Result of parsing `A1` to `A4` into minutes.

Activity	Minimum	Maximum	Missing	Values given as `HH:MM`
Working (`A1`)	0	1050	4	81
Sleeping (`A2`)	70	900	8	80
Reading (`A3`)	0	608	12	82
Dining at a restaurant (`A4`)	0	240	9	82

All four activities fall in a plausible range, and no day has a total above 1440 minutes. Parsing looks correct and no impossible values were introduced.

3.3 The age variable

The documentation describes `IKAL1` as a 9-level categorical age group, but the raw values range from 20 to 84, indicating that the dataset stores the actual ages in years instead.

We treat `IKAL1` as age in years (`age_years`) and then bin it into the age ranges of the documentation to create `age_group`. The first two bands (10 to 14 and 15 to 19) stay empty, because nobody in the sample is under 20.

3.4 Recoding the other categorical variables

The variables `pvknro`, `sp` and `ASALUE` follow the documentation exactly, so we only added readable labels (working day or weekend, male or female, city, municipality or rural area). We also checked that there are no unexpected codes or missing values in these variables

3.5 The library variable (`A5`)

`A5` should be strictly 1 (yes) or 2 (no). The raw file contains other values as well.

Table 4: Raw values of `A5` (745 rows).

Raw value	Rows
1.0 (yes)	474
2.0 (no)	209
`?` (missing marker)	42
0.0 (not documented)	17
120.0, 60.0, 420.0 (not documented)	1 each

Beyond the `?` markers, `A5` has 17 rows with the value 0.0 and three minute-like entries. Rather than guess what they mean, we followed the documentation: only 1.0 and 2.0 are recoded to "Yes" and "No", and everything else becomes missing. This makes 62 of the 745 rows (8.3%) missing for `A5`. It is a real cost, but safer than inventing an interpretation for values the documentation does not define.

3.6 Missing values

Table 5 shows the missing values in the cleaned table. The four time variables each miss at most 1.6% of their rows, and 33 rows (4.4%) miss at least one of the four.

Table 5: Missing values in the cleaned table (745 rows). Demographic variables have none.

Variable	Missing (n)	Missing (%)
`working`	4	0.54
`sleeping`	8	1.07
`reading`	12	1.61
`dining_restaurant`	9	1.21
`library_visit`	62	8.32

The amount of missing data in the time variables is small. Dropping rows is therefore fine for the analyses that need all four variables at once (correlation and PCA). For anything about a single activity, we keep all the valid data instead of discarding a whole row because of a gap in another activity. For `library_visit` we only need a yes/no proportion, so we leave out the invalid rows in that calculation and keep them everywhere else.

3.7 Outliers and inconsistencies

We checked that every time value lies between 0 and 1440 minutes, that ages are realistic, and that all codes match the documentation. No value is out of range, and the actual age range is 20 to 84.

The highest values are 1050 minutes of work (17.5 hours), 900 minutes of sleep (15 hours) and 608 minutes of reading (10 hours). These are extreme, but they are possible within one day, for example a very long shift or a day off. We keep them, because they look like real behaviour and not like data entry errors.

We did not use a statistical outlier rule such as the IQR or z-scores. These variables have many genuine zeros (someone who did not work, read or eat out that day), so such a rule would flag the large values as extreme even though zero is the most common valid answer. A physical check, can this many minutes fit in a day, is more meaningful for activity-time data. Later we use rank-based methods, which are not pulled strongly by a few large values.

3.8 Household-level table and unit of analysis

Because each household has one member, individual and household times are identical. We still built a household-day table (`df_household_day`, 745 rows) next to the person-day table (`df_clean`, 745 rows), which is the one used in Step 3.

Each person appears in up to two rows, and many tests assume independent observations. Table 6 shows, for every analysis, the unit we used and how we avoid counting a person twice.

Table 6: Unit of analysis and independence in each analysis.

Question	Unit	How we avoid counting a person twice
– Average time per activity (Section 5)	Person, within each day type	Each day type is analysed on its own, so every person appears once per estimate.
– Working day vs. weekend (Section 6.1.1)	Person	Paired test: each person is compared with themselves.
– Association between activities (Section 7)	Person, within each day type	Analysed separately by day type, so there is one row per person. Averaging a person's two days would blur the daily patterns.
– Living environment or library visit vs. activities (Sections 6.1.2 and 7.2)	Person	These are person-level variables, so each person's two days are combined into one weighted daily average (5/7 working day, 2/7 weekend).

The exploratory parts (correlation matrix and PCA in Section 4) use all person-day rows and are descriptive only, so they have no p-values.

Summary of this section

We started from 745 raw rows with mixed time formats, an undocumented age coding and a few miscoded `A5` values. We now have two clean tables, one with a row per person per diary day and one at household-day level. Missing data is small (under 2% per activity and 8.3% for library visits), and no impossible values remain.

4 Data exploration: descriptive statistics and PCA

This section describes and characterises the individuals that are present in the data (Question 1) and how people spend their time. We then use a correlation matrix and a PCA to get a first picture of how the activities relate to each other (Question 4), before the formal tests in the later sections.

4.1 Characterise the individuals that are present in the data.

We summarise the demographic variables on the person level (378 people), so that nobody is counted twice. The day type belongs to the row and not to the person, so it is not included.

Table 7: Characteristics of the 378 individuals in the sample.

Variable	Category	n	%
Sex	Female	196	51.9
	Male	182	48.1
Age group	20–24	12	3.2
	25–34	57	15.1
	35–44	64	16.9
	45–54	81	21.4
	55–64	93	24.6
	65–74	44	11.6
	75 and older	27	7.1
Living environment	City	240	63.5
	Municipality	58	15.3
	Rural area	80	21.2

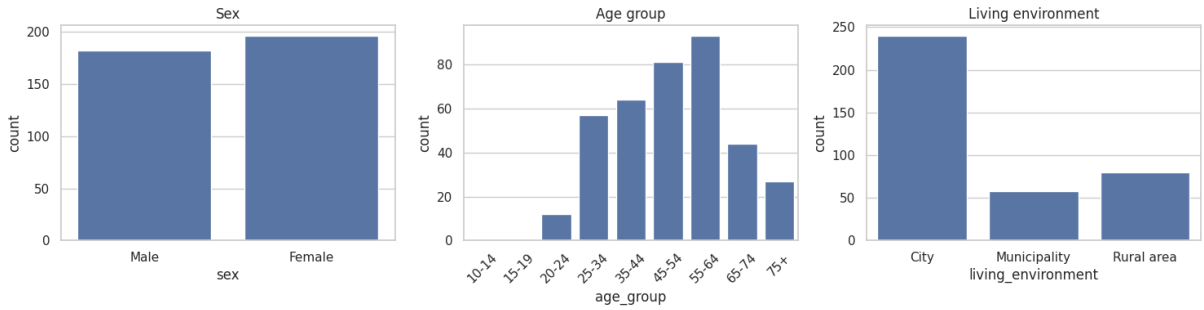


Figure 1: Number of individuals by sex, age group and living environment.

The sample has 378 people. It is close to balanced by sex (51.9% women, 48.1% men). Everyone is an adult aged 20 to 84. The largest age groups are 55–64 (24.6%) and 45–54 (21.4%), and about 19% of people are 65 or older. Most people live in cities (63.5%), while 21.2% live in rural areas and 15.3% in municipalities.

4.2 Time spent on each activity

For the four time variables we report the mean, median, standard deviation, minimum and maximum. We also report the skewness, computed on the 712 rows that have all four activities.

Table 8: Descriptive statistics of daily time per activity (minutes, person-day rows).

Activity	Mean	Median	SD	Min	Max	Mean (h)	Skewness
Working	122.1	0.0	209.0	0	1050	2.0	1.37
Sleeping	518.5	510.0	100.3	70	900	8.6	−0.10
Reading	48.0	30.0	68.1	0	608	0.8	2.92
Dining at a restaurant	48.6	0.0	58.7	0	240	0.8	0.63

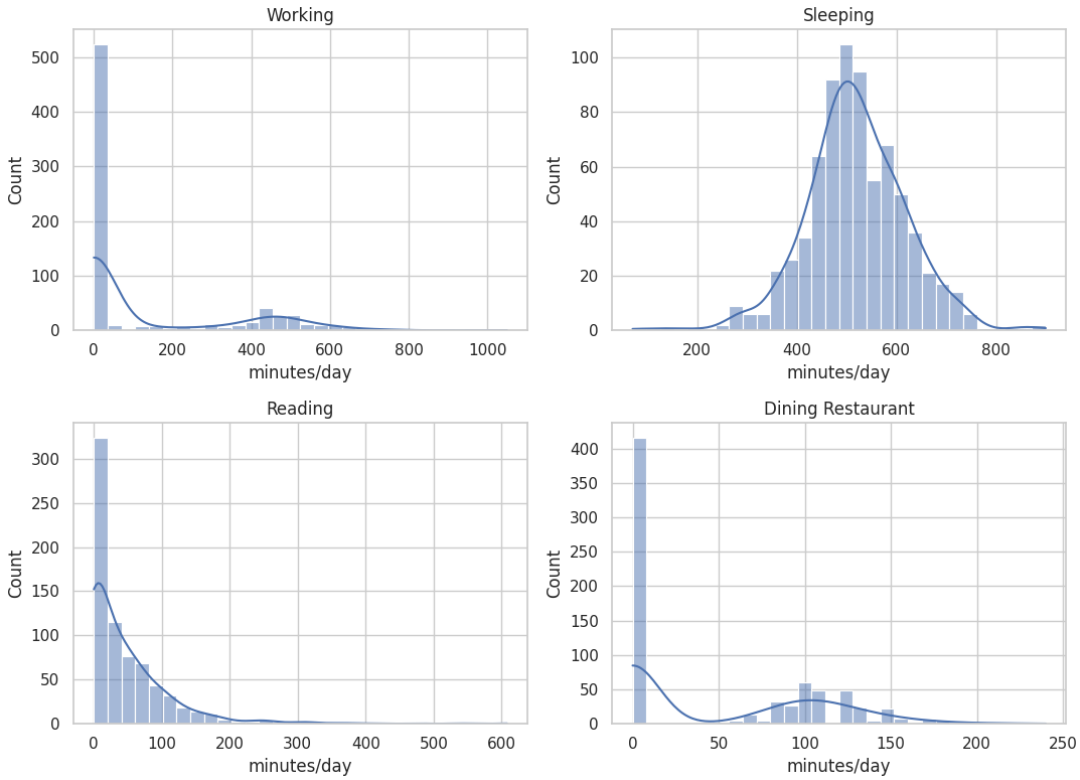


Figure 2: Distribution of daily minutes for each activity, with a kernel density curve.

Working and dining out are strongly right-skewed with a large group at 0: on many days there is no work or no restaurant visit at all. For working, the median is 0 while the mean is about 122 minutes (2 hours), so a minority of long working days pulls the average up. Sleeping looks roughly symmetric, with a mean of about 8.6 hours and a median of 8.5 hours. Reading is also right-skewed with a spike at 0 (median 30 minutes, mean 48). This skew is why we prefer rank-based methods such as Spearman correlation.

4.3 A first look at group differences

Question 3 asks whether living environment or day of the week makes a difference. We test this formally in Section 6. Figure 3 gives a first visual impression.

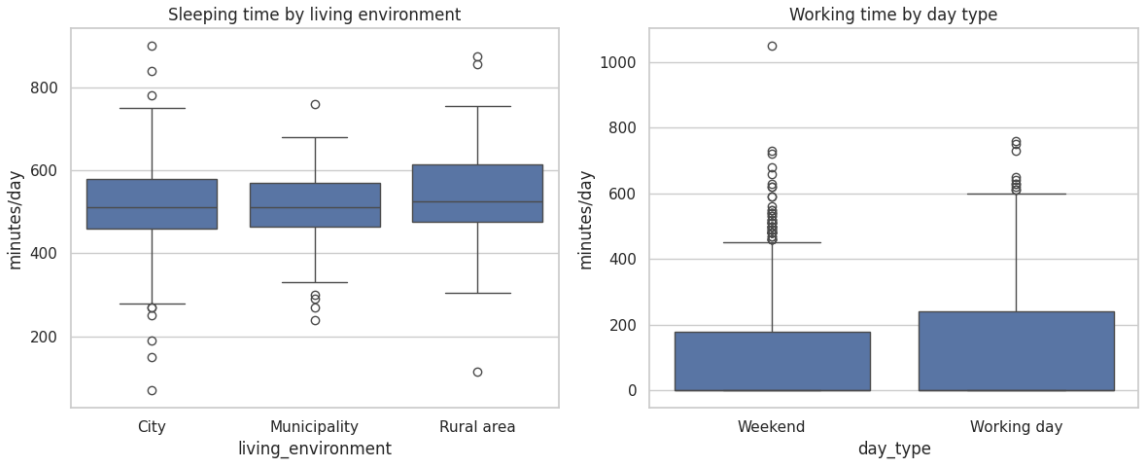


Figure 3: Left: sleeping time by living environment. Right: working time by day type.

Sleeping time is fairly stable across living environments, centred around 8 to 9 hours, with only a few low and high outliers. Working time looks very similar on both day types: working days and weekends show roughly the same mean for working hours, and most people report 0 minutes on both.

This is a surprising result, because one would expect more work on working days. In our interpretation, it comes from the mixed make-up of the sample. It includes retirees, students, and self-employed people or shift workers, whose flexible or non-traditional schedules smooth out the differences between weekdays and weekends. The data has no occupation variable, so we cannot test this explanation directly.

4.4 Library visits (`A5`)

`A5` records whether a person visited a library at least once in the past 12 months, so it is summarised as a yes/no proportion with one row per person.

Table 9: Library visits in the past 12 months (per person).

Answer	n	% of valid answers
Yes	240	68.8
No	109	31.2
No valid answer	29	—

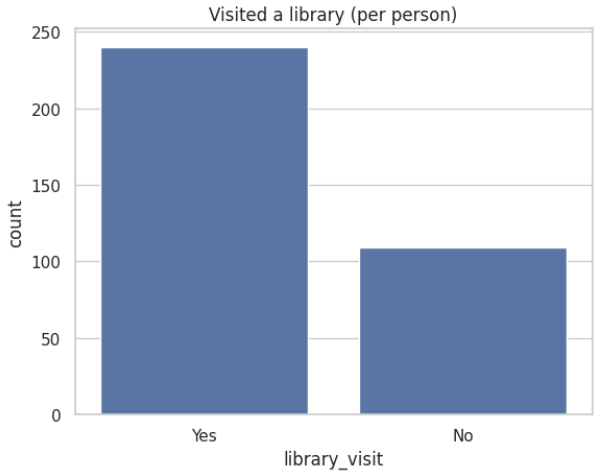


Figure 4: Number of people who visited a library in the past 12 months.

Among valid responses ($n = 349$), about 68.8% ($n = 240$) report having visited a library in the past year, while 31.2% ($n = 109$) did not. A further 29 people have no valid answer. Whether this behaviour differs across groups is tested in Section 6.

4.5 Correlation matrix between activities

As a first answer to Question 4, and as preparation for the PCA, we look at how the four activities move together. Pearson correlation assumes a roughly linear relationship and is sensitive to skew and outliers. Since the histograms showed clear right-skew with spikes at zero, we use Spearman's rank correlation, which only needs a monotonic relationship. We use the 712 rows with all four activities.

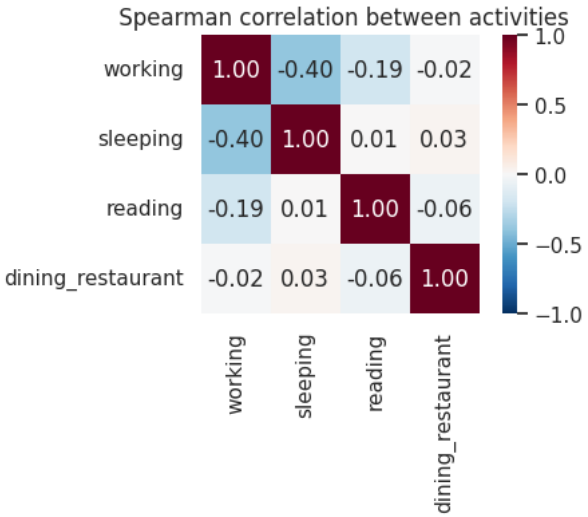


Figure 5: Spearman correlation between the four activities (712 person-day rows).

The clearest relationship is between working and sleeping: a moderate negative correlation (-0.40). Days with more work tend to have less sleep. There is also a weak negative link

between working and reading (about −0.19). The other pairs are close to 0, so the four activities are only weakly related to each other. This already suggests that a PCA will not find one strong, easily summarised component. Correlation does not show cause, so we only say that these activities move together, not that one causes the other.

Note: this exploratory correlation uses person-day rows (working days and weekends together). Since people contribute up to two rows, it shows overall patterns and not strictly independent subjects. The formal significance test is in Section 7, split by day type so that there is one row per person.

4.6 Principal Component Analysis

The PCA checks whether the four activity-time variables share a structure that can be summarised in fewer dimensions. All four variables are in minutes, but their variances differ a lot: sleeping varies in a narrow band compared with working or dining out. PCA looks for directions of maximum variance, so a variable with a large variance only because of its scale would dominate. We standardise each variable to mean 0 and variance 1, which lets every variable contribute equally. The PCA uses 712 complete rows.

Table 10: Eigenvalues and explained variance of the principal components.

Component	Eigenvalue	Explained variance (%)	Cumulative (%)
PC1	1.420	35.4	35.4
PC2	1.082	27.0	62.5
PC3	0.960	24.0	86.4
PC4	0.544	13.6	100.0

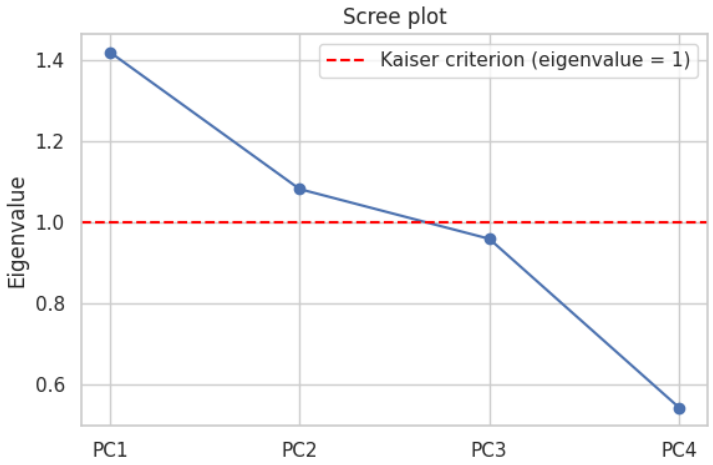


Figure 6: Scree plot of the eigenvalues. The dashed line is the Kaiser criterion (eigenvalue equal to 1).

Only PC1 (eigenvalue 1.42) and PC2 (1.08) pass the Kaiser criterion, so we keep these two. The cut is not very sharp: PC2 is barely above 1 and PC3 (0.96) is just below it. Together the two components explain about 62.5% of the variance. This is only a moderate share, which means about 38% of the variation is not shown in a two-dimensional view.

Table 11: Weights of each activity in the two retained components (eigenvector coefficients).

Activity	PC1	PC2
Working	0.718	−0.102
Sleeping	−0.649	−0.318
Reading	−0.237	0.758
Dining at a restaurant	−0.084	−0.561

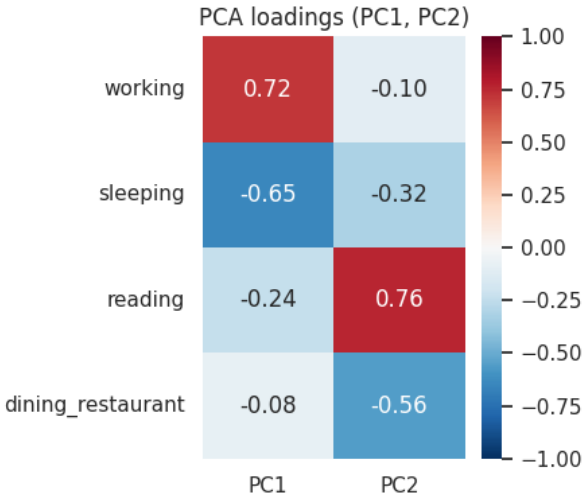


Figure 7: Heatmap of the PCA weights for PC1 and PC2.

Looking at the weights, PC1 works like a "work versus rest" axis: it contrasts working (+0.72) with sleeping (−0.65), so a high score means someone who works more and sleeps less. This is the main trade-off in the data. PC2 works more like a leisure axis: it contrasts reading (+0.76) with dining out (−0.56) and sleeping (−0.32), separating people who spend spare time reading from those who eat out. The work and rest trade-off is the strongest pattern, and the second component picks up a weaker split between kinds of leisure time. The PCA uses all 712 person-day rows, so a person can appear twice. It is a descriptive summary, not a test.

Summary of this section

The sample has 378 adults aged 20 to 84, almost balanced by sex and mostly living in cities. Work and dining out are skewed with many zeros, and sleep is about 8.6 hours a day. Work and sleep have the clearest link (more work, less sleep), and the other pairs are weak. Two principal components explain about 62.5% of the variance, which is only moderate. This step is descriptive: the formal tests come later.

5 Estimation of average daily time

This section answers Question 2: how much time do households spend on average on each activity, per day?

5.1 Confidence intervals for the activity means

We calculate the sample mean and a 95% confidence interval for each time activity, separately for working days and weekends, so that we can see how routines change.

For the intervals we use the t-based formula (Section 2). A normality test is not needed here: with about 370 observations per day type, the Central Limit Theorem makes the sample mean approximately normal, which is a fair approximation even for skewed variables such as working and dining out.

Observations within each day type are independent, as each household appears only once. Although weekdays and weekends involve the same individuals, their intervals are estimated separately here without direct comparison, which is instead addressed in Section 6.1.1.

Table 12: Average daily time per activity with 95% confidence intervals (minutes per day).

Day type	Activity	Mean	95% CI	Std. error	n
Working day	Working	122.1	[100.9, 143.3]	10.8	375
	Sleeping	515.9	[505.0, 526.7]	5.5	373
	Reading	44.7	[38.1, 51.2]	3.3	371
	Dining at a restaurant	41.5	[36.2, 46.8]	2.7	372
Weekend	Working	122.1	[100.5, 143.6]	10.9	366
	Sleeping	521.3	[511.7, 530.9]	4.9	364
	Reading	51.4	[44.0, 58.9]	3.8	362
	Dining at a restaurant	55.9	[49.3, 62.5]	3.4	364

Sleep is the largest activity: about 516 minutes (8.6 hours) on working days and 521 minutes on weekends. Reading and dining out take about 45 to 56 minutes a day, and both are a little higher on weekends.

Working days and weekends show roughly the same mean for working hours (122.1 minutes on both, with an interval of about 100 to 143 minutes). This is surprising. In our interpretation, it comes from the mixed make-up of the sample: it includes retirees, students, and self-employed people or shift workers, whose flexible or non-traditional schedules smooth out the differences between weekdays and weekends. The data has no occupation variable, so we could not check this directly.

5.2 Overall daily average, weighted by the real week

Table 12 gives the main breakdown by day type. As an extra view, we also compute one overall daily average. To reflect a real calendar week (5 working days and 2 weekend days), we combine the two means with the weights 5/7 and 2/7:

$$\bar{x}_{\text{week}} = \tfrac{5}{7}\,\bar{x}_{\text{working day}} + \tfrac{2}{7}\,\bar{x}_{\text{weekend}}.$$

A simple 50/50 average would give the weekend too much weight. This is a point estimate only, without a confidence interval.

Table 13: Weighted daily average (5/7 working day, 2/7 weekend), in minutes per day.

Activity	Working-day mean	Weekend mean	Weighted daily mean
Working	122.1	122.1	122.1
Sleeping	515.9	521.3	517.4
Reading	44.7	51.4	46.6
Dining at a restaurant	41.5	55.9	45.6

5.3 Library visits

Activity 5 is a yes/no variable, so no average can be computed. We summarise it with frequencies and proportions, with one row per person (Table 9). Most people report having visited a library in the past 12 months (about 68.8%), while about 31.2% say no. Twenty-nine people have no valid answer, and the percentages are calculated without them.

Summary of this section

We estimated the average daily time and a 95% confidence interval for each activity, separately for working days and weekends. Combining the day types with weights 5/7 and 2/7 gives one average day. About 69% of people with a valid answer visited a library in the past year. The mean working time is about 122 minutes on both day types.

6 Statistical testing: group differences

This section answers Question 3: do the type of day or the living environment make a difference in the time spent on any activity?

Before the tests, a word on multiple testing. In this section we run several tests at once, one for each activity. The more tests we run, the more likely it is that one of them looks significant by pure chance. To control this, we adjust the p-values with the False Discovery Rate method (Section 2). We apply it to each group of tests that answer the same question, and we base every decision on the adjusted p-value p_{fdr}.

6.1 Continuous activities

6.1.1 Day of the week effect

The same people are measured on a working day and on a weekend day, so the two samples are dependent and we use a paired design: each person is compared with themselves. The 11 people with only one of the two days cannot be paired and are left out of this test.

We first test normality with the Shapiro-Wilk test on the paired differences $d =$ working day $-$ weekend, supported by distribution plots. If $p \geq 0.05$ we use the paired t-test, and if $p < 0.05$ we use the Wilcoxon signed-rank test. We correct the four p-values with FDR.

- H_0: for a given activity, the type of day makes no difference to the time spent (working day and weekend are the same).
- Decision rule: if $p_{\text{fdr}} < 0.05$, we reject H_0 for that activity. If $p_{\text{fdr}} \geq 0.05$, we fail to reject H_0.

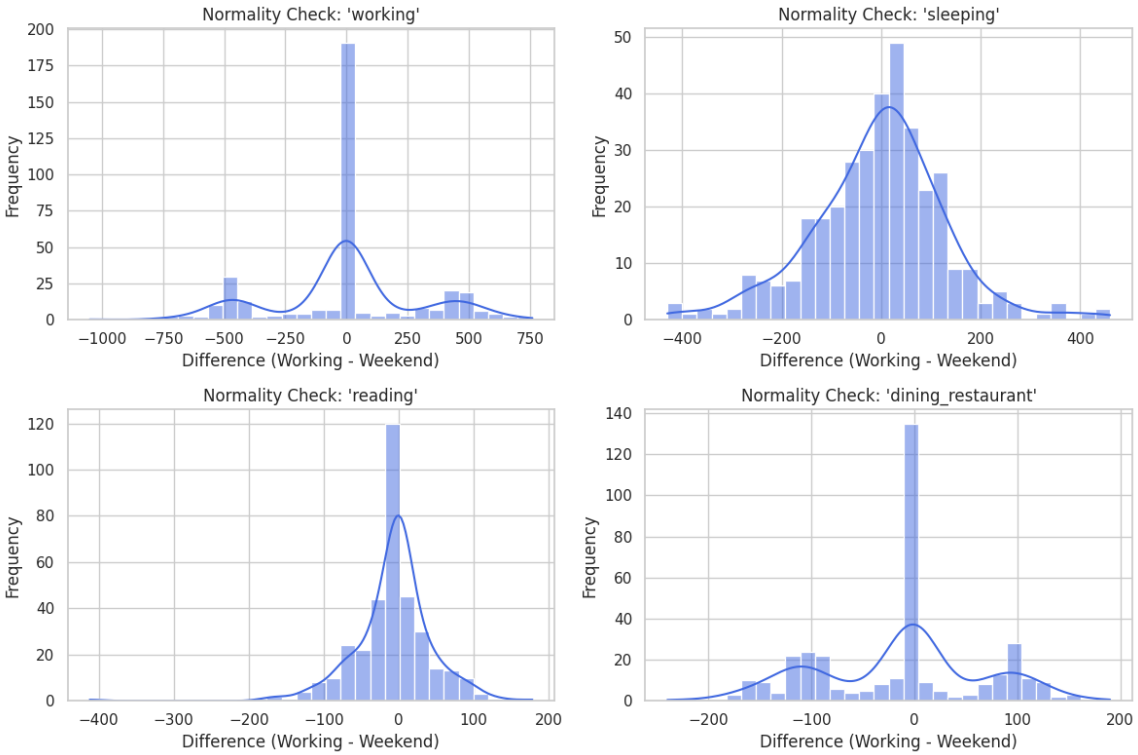


Figure 8: Distribution of the paired differences (working day minus weekend) for each activity.

Table 14: Paired tests for the day of the week effect. All four differences were non-normal, so the Wilcoxon signed-rank test was used.

Activity	Shapiro-Wilk p	Wilcoxon W	p	p_{fdr}	Significant
Working	6.95×10^{-16}	7415.5	0.6716	0.7101	No
Sleeping	1.90×10^{-5}	28646.5	0.7101	0.7101	No
Reading	9.69×10^{-15}	16094.5	0.0109	0.0217	Yes
Dining at a restaurant	1.43×10^{-10}	10161.0	0.0002	0.0008	Yes

The histograms of the paired differences are skewed with heavy tails for every activity, and the Shapiro-Wilk test agrees ($p < 0.05$ in all four cases), so we use the non-parametric Wilcoxon signed-rank test instead of the paired t-test. After FDR correction, reading ($p_{\text{fdr}} \approx 0.022$) and dining out ($p_{\text{fdr}} \approx 0.0008$) differ between day types. In both cases the average is higher on weekends (reading 51.4 vs. 44.7 minutes, dining out 55.9 vs. 41.5 minutes, from Table 12).

Working ($p_{\text{fdr}} \approx 0.71$) and sleeping ($p_{\text{fdr}} \approx 0.71$) show no significant difference. Working days and weekends have roughly the same mean for working hours (122.1 minutes on both). In our interpretation, this comes from the mixed make-up of the sample: it includes retirees, students, and self-employed people or shift workers, whose flexible or non-traditional schedules smooth

out the differences between weekdays and weekends. We could not test this directly because the data has no occupation variable.

6.1.2 Living environment effect

Living environment is a characteristic of the person, and each person has two diary days. To keep one independent observation per person, we first combine the two days into a weighted daily average (5/7 for the working day and 2/7 for the weekend day). People missing one of the two days (or an activity value) are left out for that activity, so n is a little below 378.

Next, we check the normality of each activity inside the three living environments with the Shapiro-Wilk test and boxplots. If every group looks normal we use ANOVA, otherwise the Kruskal-Wallis test.

- H_0: for a given activity, the living environment makes no difference to the time spent (city, municipality and rural area are the same).
- Decision rule: if $p_{\text{fdr}} < 0.05$, we reject H_0 for that activity. If $p_{\text{fdr}} \geq 0.05$, we fail to reject H_0, which means there is not enough evidence of a difference.

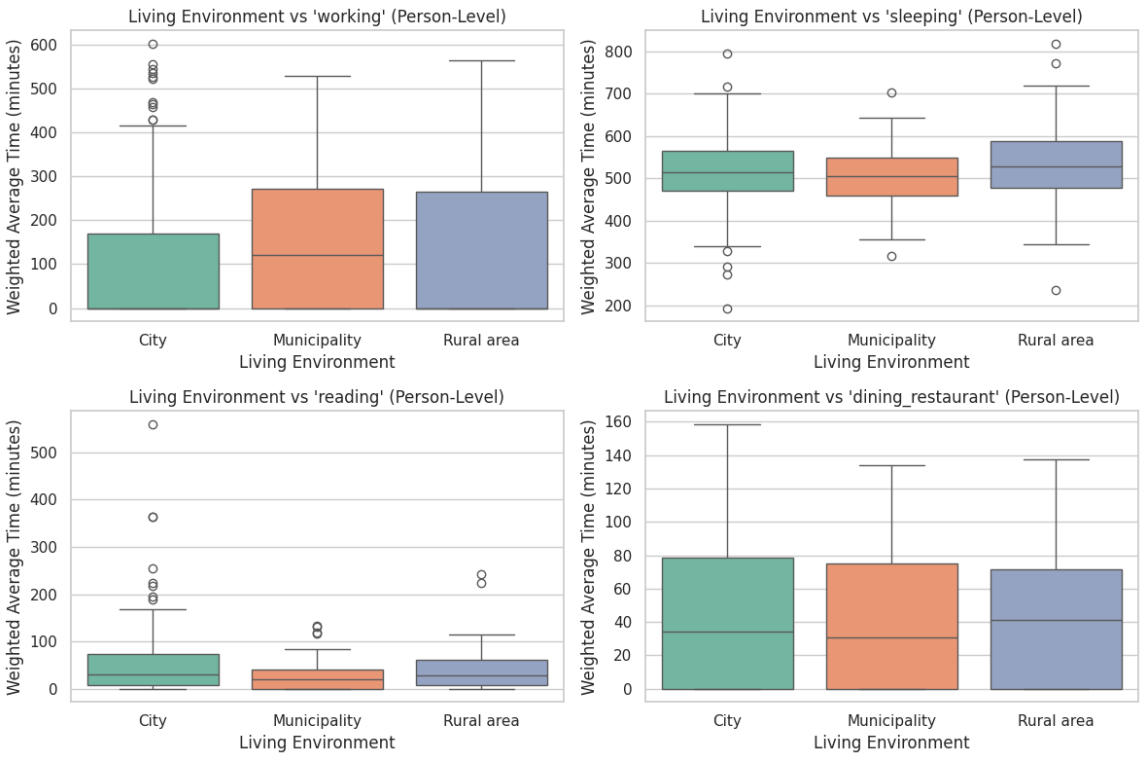


Figure 9: Weighted average daily time by living environment (one value per person).

Table 15: Kruskal-Wallis tests for living environment (person level).

Activity	n	KW	p	p_{fdr}	Significant
Working	363	2.2569	0.3235	0.4314	No
Sleeping	359	3.8219	0.1479	0.2959	No
Reading	355	6.4346	0.0401	0.1603	No
Dining at a restaurant	359	1.4420	0.4863	0.4863	No

The boxplots show different shapes across groups. Sleeping looks fairly balanced and symmetric, while working and reading are skewed. Because normality is checked in all three groups at once, a single group with heavy tails is enough to reject it, so the Kruskal-Wallis test is used for all four activities. After FDR correction, none of the activities differ significantly across living environments. Reading has a raw p-value of 0.0401, but it is not significant after correction ($p_{\text{fdr}} \approx 0.160$), and the others are clearly not significant either: working ($p_{\text{fdr}} \approx 0.43$), sleeping ($p_{\text{fdr}} \approx 0.30$) and dining out ($p_{\text{fdr}} \approx 0.49$).

We therefore fail to reject H_0. There is not enough evidence that time spent on each activity differs by living environment. This does not prove that there is no difference, only that this sample cannot show one.

6.2 Library visits and living environment

Library visit is a yes/no variable, not a time in minutes. To test whether library-visiting habits differ across living environments we use Pearson's chi-squared test of independence.

- H_0: library visits are independent of the living environment.
- H_1: there is an association between library visits and the living environment.

The library answer is a person-level attribute, so this test uses one row per person and not one row per diary day. We also do not test it against the day type: a person's 12-month library answer is the same on both of their rows, so that comparison would compare a variable with a copy of itself.

Table 16: Library visits by living environment (person level, valid answers only).

Living environment	No	Yes	Share "Yes" (%)
City	61	156	71.9
Municipality	17	40	70.2
Rural area	31	44	58.7

The chi-squared test gives $\chi^2 = 4.600$ (2 degrees of freedom) with a p-value of 0.1003. The p-value is above 0.05, so we fail to reject H_0: there is no statistically significant association between library visits and living environment. Rural areas look lower (about 59% against 70 to 72%), but with these sample sizes the gap is not large enough to rule out chance. The test assumes that expected counts are not too small. The smallest expected count is about 17.8 (municipality and "No": $57 \times 109/349$), well above the usual minimum of 5, so the test is appropriate.

Summary of this section

Reading and dining out are higher on weekends, while working and sleeping do not differ significantly between the two day types. Working days and weekends show roughly the same mean for working hours (about 122 minutes). We link this to the mixed make-up of the sample, but we could not test it. No activity differs significantly across city, municipality and rural area after FDR correction, and library visits are not significantly linked to living environment ($p = 0.10$).

7 Statistical testing: associations between activities

This section answers Question 4: which activities are associated with each other?

- H_0: for a given pair of activities, there is no monotonic association between the time spent on them (Spearman's $\rho = 0$).
- Decision rule: if $p_{\text{fdr}} < 0.05$, we reject H_0 and conclude that the two activities are associated. If $p_{\text{fdr}} \geq 0.05$, we fail to reject H_0.

7.1 Association between the time activities

Each person has two entries (a working day and a weekend day), so we separate the data by day type. This gives one row per person in each analysis, so the observations are independent, and it lets us see the relationships inside each context. We use Spearman's rank correlation because the time-use data is skewed and zero-inflated. With four activities there are six pairs per day type, so we apply the FDR correction to those six tests. Many people share the value 0, and Spearman handles these ties with average ranks.

Table 17: Spearman correlations between activities on weekends (352 complete cases of 368 rows).

Pair	ρ	p	p_{fdr}	Significant
Working and sleeping	−0.4225	< 0.0001	< 0.0001	Yes
Working and reading	−0.2168	< 0.0001	0.0001	Yes
Working and dining out	0.0248	0.6433	0.9034	No
Sleeping and reading	−0.0091	0.8642	0.9034	No
Sleeping and dining out	0.0065	0.9034	0.9034	No
Reading and dining out	−0.0954	0.0737	0.1474	No

Table 18: Spearman correlations between activities on working days (360 complete cases of 377 rows).

Pair	ρ	p	p_{fdr}	Significant
Working and sleeping	−0.3768	< 0.0001	< 0.0001	Yes
Working and reading	−0.1642	0.0018	0.0053	Yes
Working and dining out	−0.0687	0.1937	0.3874	No
Sleeping and reading	0.0164	0.7567	0.7567	No
Sleeping and dining out	0.0548	0.2996	0.4494	No
Reading and dining out	−0.0181	0.7316	0.7567	No

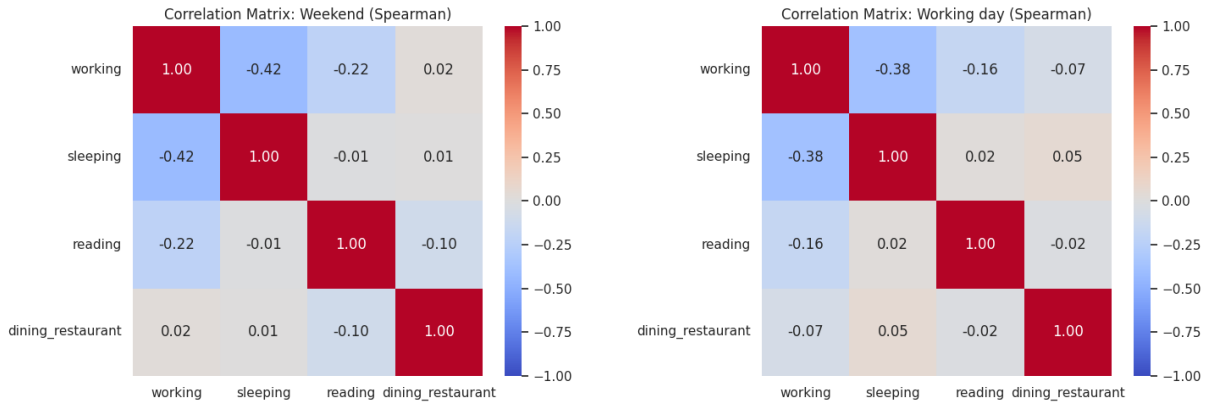


Figure 10: Spearman correlation matrices by day type. Left: weekend. Right: working day.

After FDR correction, two relationships are significant on both day types. The strongest is between working and sleeping, a moderate negative correlation on weekends ($\rho = -0.422$, $p_{\text{fdr}} < 0.001$) and on working days ($\rho = -0.378$, $p_{\text{fdr}} < 0.001$). The second is a weak negative correlation between working and reading on weekends ($\rho = -0.217$, $p_{\text{fdr}} \approx 0.0001$) and on working days ($\rho = -0.164$, $p_{\text{fdr}} \approx 0.0053$).

In other words, people who spend more time working tend to sleep less and read less, which naturally fits the constraint that a day has a limited number of hours. However, it is essential to note that correlation measures association rather than causation; these results do not imply that working directly causes lower sleep or reading times. All other variable pairs showed no statistically significant correlation after correction.

7.2 Library visits and the time activities

`A5` is asked once per person, but the diary has two rows per person. To avoid counting people twice, we use one row per person. For each activity, the two diary days are combined into one daily average (5/7 of the working day value plus 2/7 of the weekend value), and people missing one of the two days are left out for that activity. For `A5` we take the first valid answer of each person, because the answer can be missing in one row and valid in the other.

To see whether the time spent on an activity differs between people who visited a library and people who did not, we compare a continuous variable across two independent groups. The data is skewed and zero-inflated, so we use the Mann-Whitney U test. The four tests are corrected together with FDR. In total, 349 people have a valid library answer (240 yes, 109 no).

Table 19: Mann-Whitney U tests: time spent by library visit (weighted daily average, minutes).

Activity	n	Median (yes)	Median (no)	p	p_{fdr}	Significant
Working	335	14.3	0.0	0.111	0.442	No
Sleeping	330	512.9	514.3	0.814	0.906	No
Reading	329	28.6	27.4	0.701	0.906	No
Dining at a restaurant	331	34.3	34.3	0.906	0.906	No

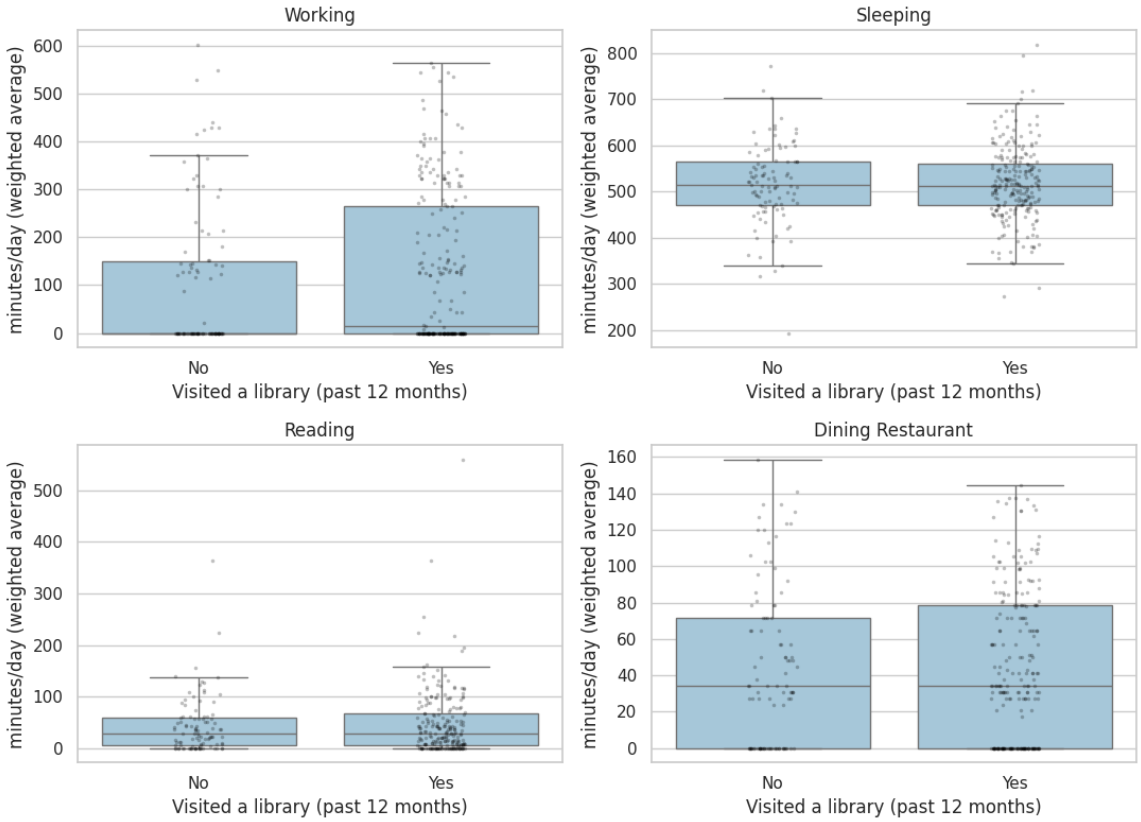


Figure 11: Time spent on each activity by library visit, with individual observations.

After FDR correction across the four activities, none of them differs significantly between people who visited a library and people who did not. All adjusted p-values are well above 0.05 (from 0.442 for working to 0.906 for sleeping, reading and dining out). Working shows a small gap in the medians (14.3 vs. 0 minutes), but it is not significant after correction. So library use is not linked to these daily activities in this sample.

Summary of this section

Working is negatively linked to sleeping (moderate) and to reading (weak), on both working days and weekends. Reading, sleeping and dining out are not significantly linked to each other. Whether a person visited a library is not significantly linked to any of the four activities.

8 Conclusions

This section answers the four questions of the brief. Each answer points to the section where the evidence is.

Question 1: Characterise the individuals that are present in the data.

The data has 378 people, each living in a different household. They are all adults aged 20 to 84, almost balanced by sex (51.9% women), and most live in cities (63.5%). About 19% are 65 or

older. Every person has up to two diary days, one working day and one weekend day (Sections 3 and 4).

Question 2: Estimate how much time on average Finnish households spend daily on each activity

- **Overall daily averages:** On an average day, Finnish households in our sample spend most of their time sleeping, about 8.6 hours (517 minutes). Working takes about 2 hours (122 minutes), reading about 47 minutes, and dining at a restaurant about 46 minutes. These figures weight a working day by 5/7 and a weekend day by 2/7 to reflect a normal week, representing point estimates without an interval.
- **Confidence intervals by day type:** The 95% confidence intervals show the precision of these averages:
  - *Sleep:* 505 to 527 minutes on working days; 512 to 531 minutes on weekends.
  - *Reading:* 38 to 51 minutes on working days; 44 to 59 minutes on weekends.
  - *Dining out:* Higher on weekends (49 to 62 minutes) than on working days (36 to 47 minutes), with non-overlapping intervals.
- **Working time interpretation:** The working time average requires careful interpretation. Over half of the recorded days contain no work, meaning the 122-minute mean comes from a subset of long days. Working days and weekends exhibit a comparable average (roughly 100 to 143 minutes), likely stemming from the sample's mixed profile (such as students, retirees, and flexible shift workers), although this could not be verified statistically.
- **Library visits:** About 69% of individuals visited a library in the past year (Section 5). Without survey weights, these findings apply strictly to our sample and cannot be automatically generalized to the entire population.

Question 3: With respect to which activities do living environments or days of week differ in Finland?

- **Days of the week:** Days of the week differ in two activities, and both differences remain significant after FDR correction. Reading and dining at a restaurant are higher on weekends (reading: 51 vs. 45 minutes; dining out: 56 vs. 42 minutes). By contrast, sleeping and working show no significant difference between working days and weekends.
- **Living environments:** For the living environment, no activity differs significantly across cities, municipalities, and rural areas. Although reading looked different initially (raw $p = 0.040$), this effect did not survive correction ($p_{\text{fdr}} \approx 0.16$).
- **Library visits:** Library visits are not significantly linked to the living environment either ($p = 0.10$), although rural areas show a somewhat lower participation rate (59% compared to about 71%).

Methodological note: Failing to find a statistically significant difference does not prove the absence of an effect, but merely indicates that our sample size is insufficient to detect one.

Question 4: Which activities are associated with each other in Finnish population?

- **Association between activities:** Working is the only activity clearly associated with others. People who work more tend to sleep less, showing a moderate link (ρ between -0.42 on weekends and -0.38 on working days), and to read less, showing a weak link (ρ between -0.22 and -0.16). Both results hold on both day types after FDR correction.
- **Interpretation and limits of correlation:** This fits the idea that the hours of a day are limited, but it does not mean that work causes the shorter sleep or reading time. Meanwhile, dining out is not significantly associated with any other activity, and neither are reading and sleeping.
- **Principal Component Analysis (PCA):** The PCA reflects the same picture: its first component contrasts working with sleeping. Furthermore, the first two components explain only 62.5% of the variance, indicating that overall links are not strong.
- **Library visits:** Library visits are not related to the time spent on any of the four activities.

A Use of AI

We used AI in this project. This appendix describes each kind of interaction, as required by the course instructions.

Table 20: Interactions with AI.

Interaction	Purpose	Outcome and checks
Review of the notebook	Second opinion on the logic and the interpretations of our notebook, and help with comments and wording.	We drove the analysis, evaluated suggestions critically, and made all final decisions based on our own reasoning rather than assuming AI outputs were correct. The code and its results remained untouched except for minor cleanups, and we re-ran the notebook to verify that the outputs stayed identical.
Explanations	Plain-language explanations of choices such as independence, the FDR correction and outliers.	It helped us understand and justify our methods, ensuring our work stayed closely aligned with what we learned in lectures.
Layout and drafting in LaTeX	Help with the structure, tables and wording of this report in Overleaf.	All numbers, tables and figures come from our notebook outputs. We read and corrected the text, and no results were invented.

How the AI guidelines were followed. The AI was not used to create or change the data. The statistical methods were chosen with the course material in mind and are explained in Section 2. The analysis code and the plots in this report come from the notebook.

**We did not save the exact conversations with the AI, so we cannot provide links to them. Instead, the table above describes how and why we used it. The interactions were mostly short exchanges in the Claude web app while working on the notebook and this report.*